

Solution Tut-2

Ans-1 :- Given, $p \equiv q \equiv T$
and $r \equiv s \equiv F$

$$(a) \quad p \vee (q \wedge r) \\ \equiv T \vee (T \wedge F) \equiv T \vee F \equiv T$$

$$(b) \quad (p \wedge (q \wedge r)) \vee \neg ((p \vee q) \wedge (r \vee s)) \\ \equiv (T \wedge (T \wedge F)) \vee \neg ((T \vee T) \wedge (F \vee F)) \\ \equiv (T \wedge F) \vee \neg (T \wedge F) \\ \equiv F \vee T \equiv T$$

Ans-2 :-

(a) $[p \wedge (p \rightarrow q)] \rightarrow q$				
p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$(p \wedge (p \rightarrow q)) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

(b) $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$						
p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	T
T	F	T	F	T	T	T
T	F	F	F	T	F	T
F	T	T	T	T	T	T
F	T	F	T	F	T	T
F	F	T	T	T	T	T
F	F	F	T	T	T	T

$$(c) (P \rightarrow q) \wedge (\neg P \rightarrow r)$$

P	q	r	$P \rightarrow q$	$\neg P$	$\neg P \rightarrow r$	$(P \rightarrow q) \wedge (\neg P \rightarrow r)$
T	T	T	T	F	T	T
T	T	F	T	F	T	T
T	F	T	F	F	T	F
T	F	F	F	F	T	F
F	T	T	T	T	F	T
F	T	F	T	T	F	T
F	F	T	T	T	F	T
F	F	F	T	T	F	F

$$(d) (P \leftrightarrow q) \vee (\neg q \leftrightarrow r)$$

P	q	r	$P \leftrightarrow q$	$\neg q$	$\neg q \leftrightarrow r$	$(P \leftrightarrow q) \vee (\neg q \leftrightarrow r)$
T	T	T	T	F	F	T
T	T	F	T	F	T	T
T	F	T	F	T	F	F
T	F	F	F	T	T	T
F	T	T	F	F	F	F
F	T	F	F	F	T	T
F	F	T	T	T	F	T
F	F	F	T	T	T	T

$$(e) (\neg P \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$$

P	q	r	$\neg P$	$\neg q$	$\neg P \leftrightarrow \neg q$	$q \leftrightarrow r$	$(\neg P \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$
T	T	T	F	F	T	T	T
T	T	F	F	F	T	F	F
T	F	T	F	T	F	T	F
T	F	F	F	T	F	F	T
F	T	T	T	F	F	T	F
F	T	F	T	F	F	F	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	F	F

Ans-3 :- When $x=1$

(a) If $x+2=3$ then $x=x+1$

If $1+2=3$ (True) then $x=1+1=2$

(b) If $\underbrace{(x+1=3)}_f$ OR $\underbrace{(2x+2=3)}_f$ then $x=x+1$
then $x=1$

(c) If $\underbrace{(2x+3=5)}_T$ AND $\underbrace{(3x+4=7)}_T$ then $x=x+1$
then $x=2$

(d) If $\underbrace{(x+1=2)}_T$ XOR $\underbrace{(x+2=3)}_T$ then $x=x+1$
then $x=1$

(e) If $\underbrace{x < 2}_T$ then $x=x+1$
then $x=2$

Ans-4 :- (a) Amar is neither rich nor happy.

(b) Virat is neither smart nor hard working.

(c) Leela will not move to Bangalore and Delhi.

Ans-5 :- (a) L.H.S = $\neg(P \vee (\neg P \wedge Q))$

$$\equiv \neg P \wedge \neg(\neg P \wedge Q)$$

(De-Morgan's law)

$$\equiv \neg P \wedge ((\neg(\neg P)) \vee \neg Q)$$

(De Morgan's law)

$$\equiv \neg P \wedge (P \vee \neg Q)$$

(Double negation law)

$$\equiv (\neg P \wedge P) \vee (\neg P \wedge \neg Q)$$

(Distribution law)

$$\equiv F \vee (\neg P \wedge \neg Q)$$

$$\equiv \neg P \wedge \neg Q$$

(Identity law)

$$\equiv R.H.S.$$

$$\begin{aligned}
 (b) \quad L.H.S. &= (P \rightarrow Q) \wedge (P \rightarrow R) \\
 &\equiv (\neg P \vee Q) \wedge (\neg P \vee R) && \text{(Implication law)} \\
 &\equiv \neg P \vee (Q \wedge R) && \text{(Distributive law)} \\
 &\equiv P \rightarrow (Q \wedge R) && \text{(Implication law)} \\
 &\equiv R.H.S.
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad L.H.S. &= (P \rightarrow R) \wedge (Q \rightarrow R) \\
 &\equiv (\neg P \vee R) \wedge (\neg Q \vee R) && \text{(Implication law)} \\
 &\equiv (\neg P \wedge \neg Q) \vee R && \text{(Distributive law)} \\
 &\equiv \neg(P \vee Q) \vee R && \text{(De Morgan's law)} \\
 &\equiv (P \vee Q) \rightarrow R && \text{(Implication law)} \\
 &\equiv R.H.S.
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad L.H.S. &= (P \rightarrow Q) \vee (P \rightarrow R) \\
 &\equiv (\neg P \vee Q) \vee (\neg P \vee R) && \text{(Implication law)} \\
 &\equiv \neg P \vee (Q \vee R) && \text{(Associative law)} \\
 &\equiv \neg P \vee (\neg P \vee Q) && \text{(Commutative law)} \\
 &\equiv (\neg P \vee \neg P) \vee Q && \text{(Associative law)} \\
 &\equiv \neg P \vee (Q \vee R) && \text{(Idempotent law)} \\
 &\equiv P \rightarrow (Q \vee R) && \text{(Implication law)} \\
 &\equiv R.H.S.
 \end{aligned}$$

$$\begin{aligned}
 (e) \quad L.H.S. &= (P \rightarrow R) \vee (Q \rightarrow R) \\
 &\equiv (\neg P \vee R) \vee (\neg Q \vee R) && \text{(Implication law)} \\
 &\equiv \neg P \vee (R \vee \neg Q) && \text{(Associative law)} \\
 &\equiv \neg P \vee (\neg Q \vee R) && \text{(Commutative law)} \\
 &\equiv (\neg P \vee \neg Q) \vee (R \vee R) && \text{(Associative law)} \\
 &\equiv \neg(P \wedge Q) \vee R && \text{(Idempotent law and De Morgan's law)} \\
 &\equiv (P \wedge Q) \rightarrow R && \text{(Implication law)} \\
 &\equiv R.H.S.
 \end{aligned}$$

$$\begin{aligned}
 (f) \quad L.H.S. &= \neg P \rightarrow (Q \rightarrow R) \\
 &\equiv \neg(\neg P) \vee (Q \rightarrow R) && \text{(Implication law)} \\
 &\equiv P \vee (\neg Q \vee R) && \text{(Double negation and implication law)} \\
 &\equiv (P \vee \neg Q) \vee R && \text{(Associative law)} \\
 &\equiv (\neg Q \vee P) \vee R && \text{(Commutative law)} \\
 &\equiv \neg Q \vee (P \vee R) && \text{(Associative law)} \\
 &\equiv Q \rightarrow (P \vee R) && \text{(Implication law)} \\
 &\equiv R.H.S.
 \end{aligned}$$

$$\begin{aligned}
 (g) \quad L.H.S. &= P \leftrightarrow Q \\
 &\equiv (P \rightarrow Q) \wedge (Q \rightarrow P) && \text{(Biconditional law)} \\
 &\equiv (\neg Q \rightarrow \neg P) \wedge (\neg P \rightarrow \neg Q) && \text{(Contrapositive law)} \\
 &\equiv (\neg P \rightarrow \neg Q) \wedge (\neg Q \rightarrow \neg P) && \text{(Commutative law)} \\
 &\equiv \neg P \leftrightarrow \neg Q && \text{(Biconditional law)} \\
 &\equiv R.H.S.
 \end{aligned}$$

$$\begin{aligned}
 (h) \quad L.H.S. &= P \leftrightarrow Q \\
 &\equiv (P \rightarrow Q) \wedge (Q \rightarrow P) && \text{(Biconditional law)} \\
 &\equiv (\neg P \vee Q) \wedge (\neg Q \vee P) && \text{(Implication law)} \\
 &\equiv (\neg P \wedge (\neg Q \vee P)) \vee (Q \wedge (\neg Q \vee P)) && \text{(Distributive law)} \\
 &\equiv ((\neg P \wedge \neg Q) \vee (\neg P \wedge P)) \vee ((Q \wedge \neg Q) \vee (Q \wedge P)) && \text{(Distributive law)} \\
 &\equiv ((\neg P \wedge \neg Q) \vee F) \vee (F \vee (Q \wedge P)) \\
 &\equiv (\neg P \wedge \neg Q) \vee (Q \wedge P) && \text{(Identity law)} \\
 &\equiv (\neg P \wedge \neg Q) \vee (P \wedge Q) && \text{(Commutative law)} \\
 &\equiv (P \wedge Q) \vee (\neg P \wedge \neg Q) && \text{(Commutative law)} \\
 &\equiv R.H.S.
 \end{aligned}$$